# PONMATHISH T

# DAY - 05

**10 Marks**

**1. What is Ethernet?**

**Ethernet** is a widely used networking technology for connecting devices in a **Local Area Network (LAN)**. It defines how devices communicate and exchange data over a wired network.

**Main points:**

1. Ethernet is mainly used for **LAN communication**.
2. It uses network cables such as **twisted-pair copper and fiber-optic cables**.
3. Ethernet operates primarily at the **Physical Layer (Layer 1)** and **Data Link Layer (Layer 2)** of the OSI model.
4. It uses **MAC addresses** to identify network interfaces.
5. Ethernet data is transmitted in the form of **frames**.
6. Ethernet switches use MAC addresses to forward frames.
7. Common Ethernet standards include **Fast Ethernet, Gigabit Ethernet, and 10 Gigabit Ethernet**.
8. It provides reliable and high-speed communication between network devices.
9. Ethernet is commonly used to connect computers, switches, routers, printers, and servers.
10. It is widely used in homes, offices, data centers, and enterprise networks.

**Example:** A computer connected to a network switch using an Ethernet cable is using Ethernet technology.

**Conclusion:** Ethernet is an important LAN technology that enables devices to communicate efficiently using frames and MAC addresses.

**2. What is an Ethernet Frame?**

An **Ethernet frame** is the basic unit of data transmitted over an Ethernet network. It contains the actual data along with addressing and control information required for delivery.

**Main parts of an Ethernet frame include:**

1. **Preamble** – Helps synchronize the receiver.
2. **Start Frame Delimiter (SFD)** – Indicates the beginning of the frame.

3. **Destination MAC Address** – Identifies the receiving network interface.
4. **Source MAC Address** – Identifies the sending network interface.
5. **EtherType/Length** – Identifies the type of network-layer protocol or payload length.
6. **Data/Payload** – Contains the data being transported.
7. **FCS (Frame Check Sequence)** – Helps detect errors in the received frame.
8. Frames are transmitted between devices on an Ethernet network.
9. Switches examine the **destination MAC address** to decide where to forward frames.
10. Ethernet frames allow data to be delivered efficiently across a local network.

**Simple structure:**

**Destination MAC → Source MAC → Type/Length → Data → FCS**

**Conclusion:** An Ethernet frame packages data with MAC addressing and error-detection information so it can be transmitted across an Ethernet network.

**3. What is a MAC Address?**

A **MAC (Media Access Control) address** is a hardware-level network identifier associated with a network interface. It is primarily used for communication within a **local network**.

**Main points:**

1. MAC addresses operate primarily at **Layer 2 – Data Link Layer**.
2. They identify network interfaces on Ethernet and other Layer 2 networks.
3. A traditional MAC address is **48 bits (6 bytes)** long.
4. It is usually written in hexadecimal format.
5. Example: **58-E4-EB-41-04-F2**.
6. Ethernet frames contain **source and destination MAC addresses**.
7. Switches learn MAC addresses from incoming frames.
8. Switches store learned addresses in a **MAC address table**.
9. MAC addresses help switches determine which port should receive a frame.

10. MAC addresses are different from IP addresses, which are used for logical addressing and routing.

**Example:**

MAC Address: 58-E4-EB-41-04-F2

**Conclusion:** A MAC address provides Layer 2 identification for a network interface and plays an important role in Ethernet frame delivery and network switching.

**4. What does MAC stand for?**

MAC stands for **Media Access Control**.

1. MAC is associated with the **Data Link Layer (Layer 2)**.
2. A MAC address identifies a network interface on a local network.
3. It is commonly used in Ethernet and Wi-Fi networks.
4. A traditional MAC address is **48 bits (6 bytes)** long.
5. It is usually written in hexadecimal format.
6. Example: 58-E4-EB-41-04-F2.
7. Ethernet frames contain source and destination MAC addresses.
8. Switches use MAC addresses to forward frames.
9. Switches learn MAC addresses from incoming frames.
10. MAC addressing helps devices communicate within a local network.

**5. Difference Between a MAC Address and an IP Address**

A **MAC address** is used for Layer 2 communication, while an **IP address** is used for Layer 3 communication and routing.

| **Feature** | **MAC Address** | **IP Address** |
|---|---|---|
| Full form | Media Access Control | Internet Protocol |
| OSI Layer | Layer 2 – Data Link | Layer 3 – Network |

| Feature | MAC Address | IP Address |
|---|---|---|
| Purpose | Identifies a network interface locally | Identifies a device/interface logically for communication and routing |
| Used by | Switches | Routers |
| Format | Hexadecimal | IPv4 or IPv6 |
| Example | 58-E4-EB-41-04-F2 | 192.168.1.10 |
| Scope | Mainly local network | Can be used across interconnected networks |
| Changes | Usually stable, but can be changed/spoofed | Can change depending on the network |

**Conclusion:** MAC addresses are mainly used for **local frame delivery**, while IP addresses are used for **logical addressing and routing between networks**.

**6. What is a MAC Address Table?**

A **MAC Address Table** is a table maintained by an Ethernet switch that records **which MAC addresses are reachable through which switch ports**.

**Main points:**

1. It is also called a **CAM (Content Addressable Memory) table**.
2. It contains learned **MAC addresses**.
3. It associates each MAC address with a switch port.
4. A switch learns the **source MAC address** from incoming frames.
5. The switch stores the learned information in its MAC address table.
6. When a frame arrives, the switch checks the destination MAC address.
7. If the destination MAC is known, the switch forwards the frame through the appropriate port.
8. If the destination MAC is unknown, the switch may **flood** the frame out other relevant ports.
9. MAC table entries can eventually be removed if they are not used for a period of time.

10. MAC learning improves network efficiency by reducing unnecessary frame forwarding.

**Example:**

| MAC Address | Switch Port |
|---|---|
| AA:BB:CC:11:22:33 | Port 1 |
| 58:E4:EB:41:04:F2 | Port 5 |

**Conclusion:** A MAC Address Table helps a switch determine **where to forward Ethernet frames**, making local network communication more efficient.

**7. How Does a Switch Learn MAC Addresses?**

A network switch learns MAC addresses automatically by examining the **source MAC address** of incoming Ethernet frames.

1. A device sends an Ethernet frame to the switch.
2. The switch receives the frame on one of its ports.
3. The switch reads the **source MAC address** in the frame.
4. It records that MAC address in its **MAC Address Table**.
5. The switch associates the MAC address with the port where the frame arrived.
6. If the MAC address already exists, the switch can update its associated port information.
7. When another frame arrives, the switch checks its MAC Address Table.
8. If the destination MAC address is known, it forwards the frame to the appropriate port.
9. If the destination MAC address is unknown, the switch may flood the frame.
10. Learned entries can be removed after a period of inactivity.

**Example:**
If a frame arrives on **Port 3** with source MAC AA:BB:CC:11:22:33, the switch learns:

AA:BB:CC:11:22:33 → Port 3

**8. What is a Source MAC Address?**

A **source MAC address** is the MAC address of the **network interface that sent an Ethernet frame**.

1. It identifies the sending device/interface at Layer 2.
2. It is found in the Ethernet frame header.
3. The switch reads the source MAC address when receiving a frame.
4. The switch uses it to learn the device's location.
5. It helps build the switch's MAC Address Table.
6. It is normally different from the destination MAC address.
7. It is used for local Ethernet communication.
8. A MAC address is typically 48 bits long.
9. Example: 58-E4-EB-41-04-F2.
10. The source MAC tells the receiving network device **where the frame came from**.

**Example:**
If Computer A sends a frame to Computer B, Computer A's MAC address is the **source MAC address**.

**9. What is a Destination MAC Address?**

A **destination MAC address** is the MAC address of the **network interface that an Ethernet frame is intended to reach on the local network**.

1. It is contained in the Ethernet frame header.
2. It identifies the intended Layer 2 destination.
3. A switch reads the destination MAC address.
4. The switch checks its MAC Address Table.
5. If the address is known, the switch forwards the frame to the appropriate port.
6. If the address is unknown, the switch may flood the frame.
7. A destination can also be a broadcast or multicast MAC address.
8. Example of a unicast MAC address: AA-BB-CC-11-22-33.
9. It is important for local Ethernet frame delivery.
10. It tells the switch **where the frame should be delivered**.

**Example:**
If Computer A sends a frame to Computer B, Computer B's MAC address is the **destination MAC address**.

**10. What is an Unknown Unicast Frame?**

An **unknown unicast frame** is an Ethernet frame whose **destination MAC address is not present in the switch's MAC address table**.

1. The frame has a specific destination MAC address.
2. The switch receives the frame.
3. The switch checks its MAC Address Table.
4. The destination MAC address is not found.
5. Therefore, the switch does not know which port leads to the destination.
6. The switch normally **floods** the frame out all appropriate ports in the same VLAN, except the port where it received the frame.
7. The destination device may then receive the frame.
8. The switch can learn the destination's MAC address when that device sends a frame back.
9. After learning it, future frames can be forwarded directly.
10. Unknown unicast flooding can temporarily increase network traffic.

**Example:**
If the switch receives a frame for MAC AA:BB:CC:11:22:33, but this MAC is not in its table, the frame is treated as an **unknown unicast**.

**11. What is Flooding?**

**Flooding** is the process in which a switch sends an Ethernet frame out **multiple appropriate ports** because it does not know the specific destination port, or because the frame is intended for multiple devices.

1. Flooding is commonly used for **unknown unicast** frames.
2. The switch checks its MAC Address Table.
3. If the destination MAC is unknown, the switch floods the frame.
4. The frame is sent through other ports in the same VLAN.

5. The frame is not normally sent back through the port where it arrived.
6. Broadcast frames are also flooded within the relevant VLAN.
7. Flooding helps ensure that the intended device can receive the frame.
8. Once the destination MAC is learned, unnecessary flooding can be reduced.
9. Excessive flooding can increase network traffic.
10. Switches use MAC learning to minimize unnecessary flooding.

**Example:**
If a switch does not know where a destination MAC address is located, it may send the frame through all other ports in that VLAN.

**12. What is Known Unicast Forwarding?**

**Known unicast forwarding** occurs when a switch knows the destination MAC address and its corresponding port in the **MAC Address Table**.

1. A switch receives an Ethernet frame.
2. It reads the destination MAC address.
3. It searches the MAC Address Table.
4. The destination MAC address is found.
5. The switch identifies the correct outgoing port.
6. It forwards the frame only through that appropriate port.
7. It does not normally flood the frame to other ports.
8. This reduces unnecessary network traffic.
9. It improves network efficiency and performance.
10. MAC learning enables switches to perform known unicast forwarding.

**Example:**

Destination MAC → AA:BB:CC:11:22:33

MAC Table → Port 4

Action → Forward frame to Port 4

**Quick Difference**

| Type | Destination MAC known? | Switch action |
|---|---|---|
| **Unknown Unicast** | No | Floods the frame |
| **Known Unicast** | Yes | Forwards to the specific port |

**13. At Which OSI Layer Does Ethernet Switching Primarily Operate?**

Ethernet switching primarily operates at **Layer 2 – Data Link Layer** of the OSI model.

1. The Data Link Layer is **Layer 2** of the OSI model.
2. Ethernet frames operate at this layer.
3. Ethernet switches primarily use **MAC addresses** to forward frames.
4. A switch learns source MAC addresses from incoming frames.
5. It stores learned addresses in a **MAC Address Table**.
6. When a frame arrives, the switch checks the destination MAC address.
7. If the destination is known, the switch forwards the frame to the appropriate port.
8. If the destination is unknown, the switch may flood the frame.
9. Ethernet switching helps devices communicate efficiently within a LAN.
10. Some advanced switches also perform Layer 3 routing, but basic Ethernet switching is primarily a **Layer 2 function**.

**Answer in short: Data Link Layer – Layer 2**

**14. Why Does a Switch Maintain a MAC Address Table?**

A switch maintains a **MAC Address Table** to determine where connected devices are located and to **forward Ethernet frames to the correct port**.

1. It stores learned source MAC addresses.
2. It associates each MAC address with a switch port.
3. It helps the switch identify where a device is connected.
4. The switch checks the destination MAC address of incoming frames.
5. If the destination is known, the switch forwards the frame to the correct port.

6. It avoids sending known unicast frames to every port.
7. This reduces unnecessary network traffic.
8. It improves network performance and efficiency.
9. MAC table entries can change when devices move between ports.
10. Entries can also expire after a period of inactivity.

**Example:**

AA:BB:CC:11:22:33 → Port 3

This tells the switch that the device with that MAC address can be reached through **Port 3**.

**15. Why is Ethernet Knowledge Important in Cybersecurity?**

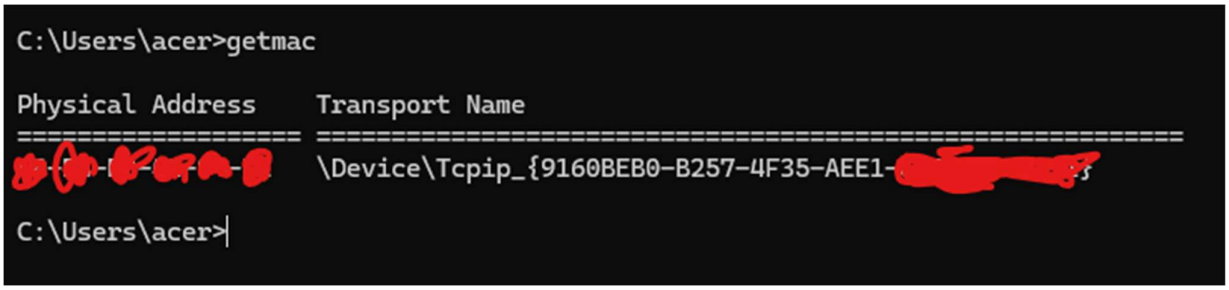
Understanding Ethernet is important in cybersecurity because many attacks, monitoring activities, and security investigations involve **local network traffic and Ethernet frames**.

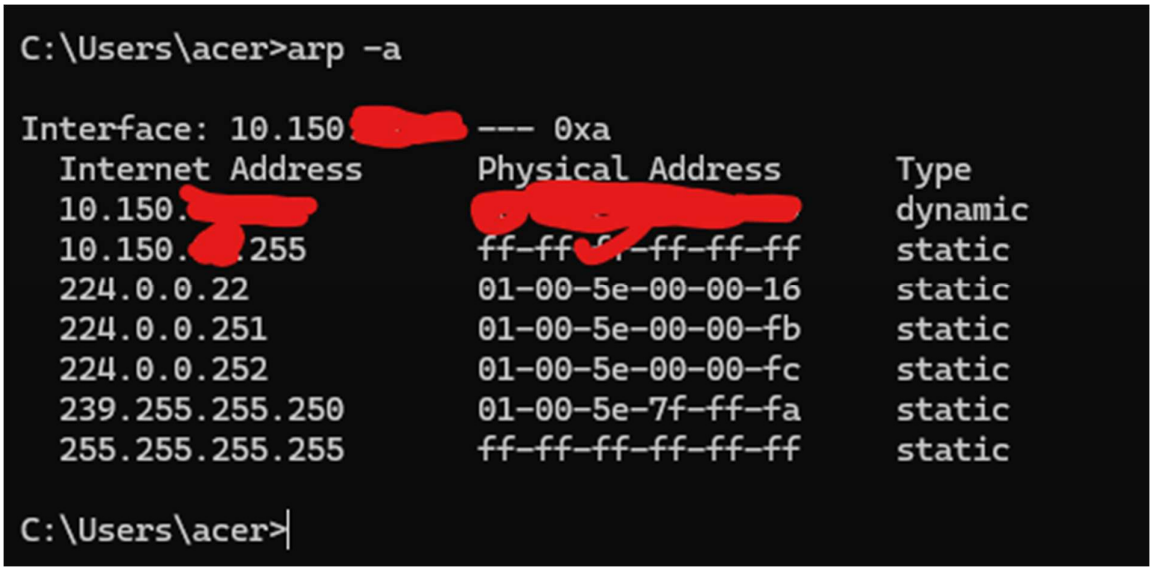
1. Helps cybersecurity professionals understand how devices communicate on a LAN.
2. Helps understand **MAC addresses and Ethernet frames**.
3. Helps analyze network traffic using tools such as Wireshark.
4. Helps detect suspicious or unusual network activity.
5. Helps understand **MAC spoofing** attacks.
6. Helps investigate unauthorized devices connected to a network.
7. Helps understand switch behavior and MAC address tables.
8. Helps identify issues related to ARP and local network communication.
9. Helps with network monitoring and incident response.
10. Provides a foundation for implementing effective network security controls.

**Conclusion:** Ethernet knowledge helps cybersecurity professionals **monitor, troubleshoot, investigate, and protect local networks from attacks**.

**PRACTICAL IDENTIFICATION**

| Details | Answer |
|---|---|
| **Network Adapter Name** | Wi-Fi – Realtek 8821CE Wireless LAN 802.11ac PCI-E NIC |
| **MAC Address** | 37-F7-BZ-51-02-E2 |
| **IPv4 Address** | 10.150.34.91 |
| **Default Gateway** | 10.150.34.120 |



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Microsoft Windows [Version [redacted].0.26100.7623]
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C:\Users\[redacted]>ipconfig/all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : [redacted]
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Wireless LAN adapter Local Area Connection* 8:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : [redacted]-E4-[redacted]-41-[redacted]
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : [redacted]-E4-[redacted]-41-[redacted]
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek 8821CE Wireless LAN 802.11ac PCI-E NIC
   Physical Address. . . . . . . . . : [redacted]-E4-[redacted]-41-[redacted]
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : [redacted]:40f4:[redacted]:23cc:[redacted]:e17e:[redacted]:2d[redacted](Preferred)
   Temporary IPv6 Address. . . . . . : [redacted]:40f4:[redacted]:23cc:[redacted]:656a:[redacted]:64[redacted](Preferred)
   Link-local IPv6 Address . . . . . : [redacted]::88e0:[redacted]:5735[redacted](Preferred)
   IPv4 Address. . . . . . . . . . . : 10.150.[redacted](Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 11 October 2026 06:24:48
   Lease Expires . . . . . . . . . . : 11 October 2026 10:26:14
   Default Gateway . . . . . . . . . : [redacted]:bc8d:[redacted]:feb1:[redacted]
                                       10.150.[redacted]
   DHCP Server . . . . . . . . . . . : 10.150.[redacted]
   DHCPv6 IAID . . . . . . . . . . . : 173597931
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-A0-86-E6-F8-E4-3B-47-B7-B7
   DNS Servers . . . . . . . . . . . : 10.150.[redacted]
                                       [redacted]:40f4:[redacted]:23c[redacted]
   NetBIOS over Tcpip. . . . . . . . : Enabled

C:\Users\[redacted]>
```